

生态研究前沿速览_2026-04-25

生态研究前沿速览

2026-04-25

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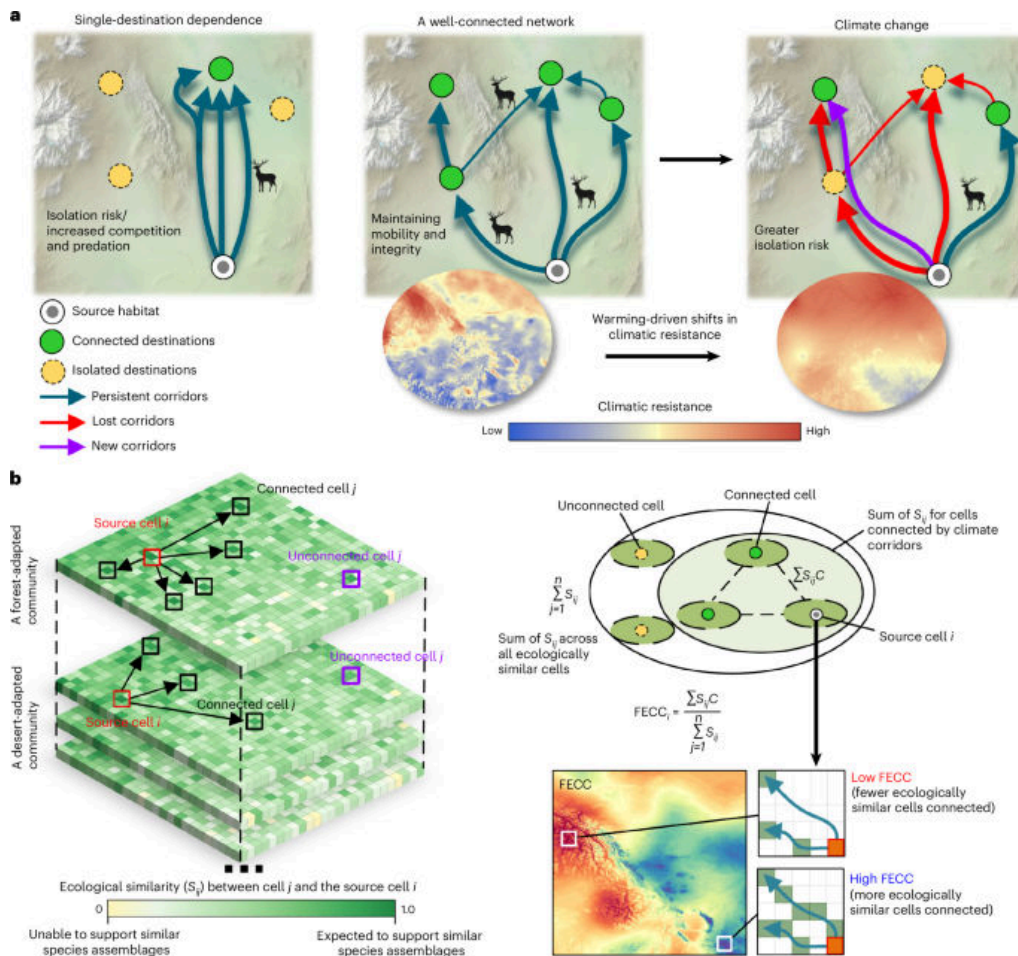
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生态学核心

1. Flexible photosynthesis at the end Permian
(二叠纪末期植物光合作用策略的灵活性)

Nature Ecology & Evolution | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Ecology & Evolution, Published online: 20 April 2026; doi:10.1038/s41559-026-03052-y The few plants that thrived after the Earth’s greatest mass extinction used a specialized physiology, known as crassulacean acid metabolism photosynthesis, that is adapted to extremely dry environments.

中文摘要

研究围绕“二叠纪末期植物光合作用策略的灵活性”展开。英文摘要要点为：The few plants that thrived after the Earth’s greatest mass extinction used a specialized physiology, known as crassulacean acid metabolism photosynthesis, that is adapted to extremely dry environments. 该条目发表于 Nature Ecology & Evolution，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41559-026-03052-y>

生态学核心

2. CAM photosynthesis may have conferred an advantage during the Permian–Triassic mass extinction event

(CAM 光合作用可能帮助植物熬过二叠纪—三叠纪大灭绝)

Nature Ecology & Evolution | 2026-04-20 | 【Latest】

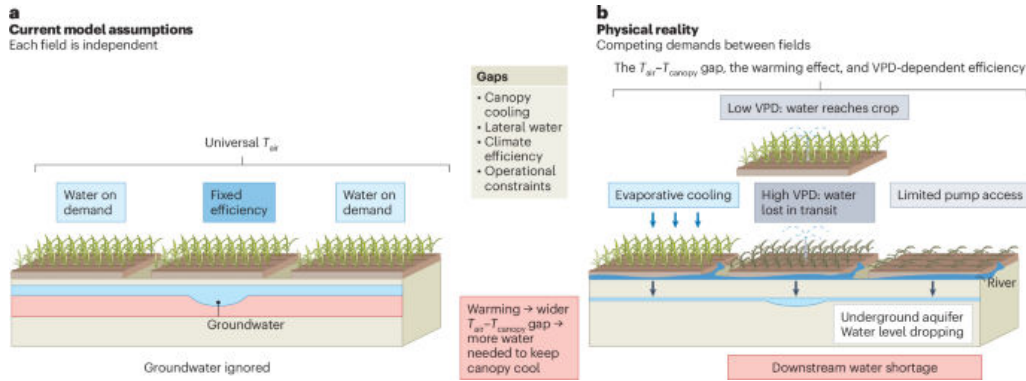


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Ecology & Evolution, Published online: 20 April 2026; doi:10.1038/s41559-026-03026-0 Morphological and stable isotope analysis of Early Triassic lycopphyte leaves suggest they were similar to extant Isoetales and thus may have made use of crassulacean acid metabolism photosynthesis—a trait that may have been advantageous during the Permian–Triassic mass extinction.

中文摘要

研究通过早三叠世石松类叶片的形态和稳定同位素分析发现，这些植物可能类似现代水韭目植物，并采用景天酸代谢（CAM）光合作用。这种更适应极端干旱与环境胁迫的生理策略，可能解释了部分植物在二叠纪—三叠纪大灭绝后仍能存活并扩张。

深度解读

这项研究把古植物学、形态证据和稳定同位素线索拼到了一起，给出了一个很有说服力的解释：并不是所有植物都被动承受大灭绝，某些类群可能靠更灵活的光合策略挺了过去。CAM 通常与干旱、高温和水分利用效率提升相关，因此它在极端气候背景下的优势非常合理。更重要的是，这篇文章提醒我们，大灭绝后的生态重建不只是‘谁死了谁活着’，还涉及生理机制层面的筛选。对理解植物如何应对未来极端气候，它也提供了深时尺度上的启示。

DOI: <https://doi.org/10.1038/s41559-026-03026-0>

环境与气候

3. Ocean warming weakens the sea–land breeze in coastal megacities

(海洋变暖会削弱沿海特大城市的海陆风降温效应)

Nature Climate Change | 2026-04-17 | 【Latest】

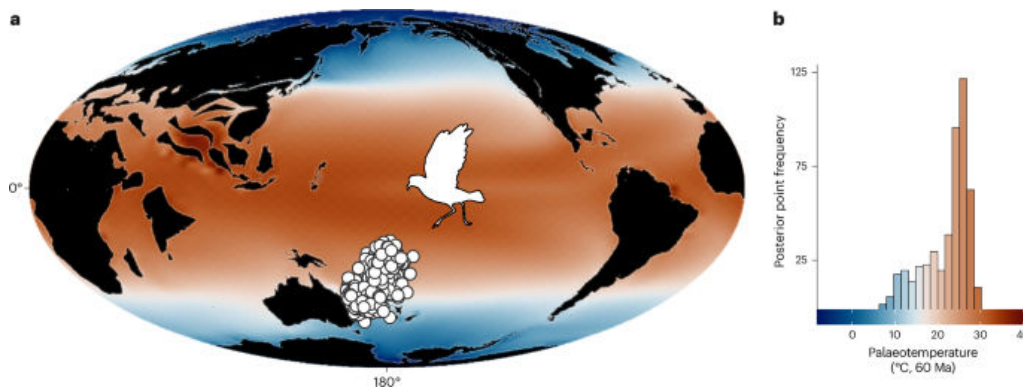


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Climate Change, Published online: 17 April 2026; doi:10.1038/s41558-026-02618-9The sea–land breeze acts to counter urban heat in many coastal cities. Here the authors simulate how this circulation changes with warming ocean water, showing that it decreases in most of them, adding heat stress to urban areas.

中文摘要

研究模拟发现，随着近海海水升温，许多沿海特大城市用于缓解城市热岛的海陆风环流将减弱，从而降低自然通风与降温能力，并进一步加剧城市热暴露风险。

深度解读

很多沿海城市把海风当成‘天然空调’，但这篇文章说明这套缓冲机制本身也会被气候变化削弱。问题不只是城市更热，而是原本能够带走热量的近海—陆地环流也在变弱，导致热风险被双重放大。对城市适应来说，这意味着单靠滨海区位并不足以抵消高温，需要把通风廊道、建筑布局 and 高温应急系统一起纳入规划。对上海、广州、深圳这类超大沿海城市尤其有现实意义。

DOI: <https://doi.org/10.1038/s41558-026-02618-9>

植物与农业

4. Designing where to crop and where to graze: a spatial approach toward sustainable farming

(用空间优化重新划定种植区与放牧区以推动可持续农业)

Agronomy for Sustainable Development | 2026-04-14 | 【Latest】



图：Graphical abstract / article figure source: <https://static-content.springer.com/image/art%3A10.1007%2Fs13593-026-01096-9>

Abstract

The world's growing population raises concerns about future food security. At the same time, environmental challenges such as climate change, biodiversity loss, and the depletion of natural resources must be addressed, calling for a transformation of agriculture. This need is exacerbated by the limited availability of land suitable for arable farming. In this study, we examined the potential of agricultural land for arable farming and grassland use in Switzerland, under the premise that agricultural land use should align with the biophysical and environmental capacity of each location. In an iterative co-design process with scientists and public authorities, we elaborated three scenarios for agricultural transformation, which progressively incorporated (i) biophysical constraints (soil, cl

中文摘要

研究以瑞士为例，依据不同地块的生物物理条件与环境承载力，评估哪些农业用地更适合耕作、哪些更适合草地与放牧。结果表明，将土地利用与地点适宜性更精确匹配，有望在粮食安全、资源效率和生态环境之间取得更优平衡。

深度解读

这篇文章的价值在于，它不再把农业问题理解成‘多生产一点就行’，而是追问：不同地方究竟最适合做什么。把耕作放在真正适宜耕作的地块，把放牧留给更适合草地利用的区域，本质上是在做农业空间重构。这样的思路既有助于减少低效投入，也能减轻土壤退化、养分流失和生境压力。对山地农业、草牧交错带和国土空间优化来说，这是一种很值得推广的方法框架。

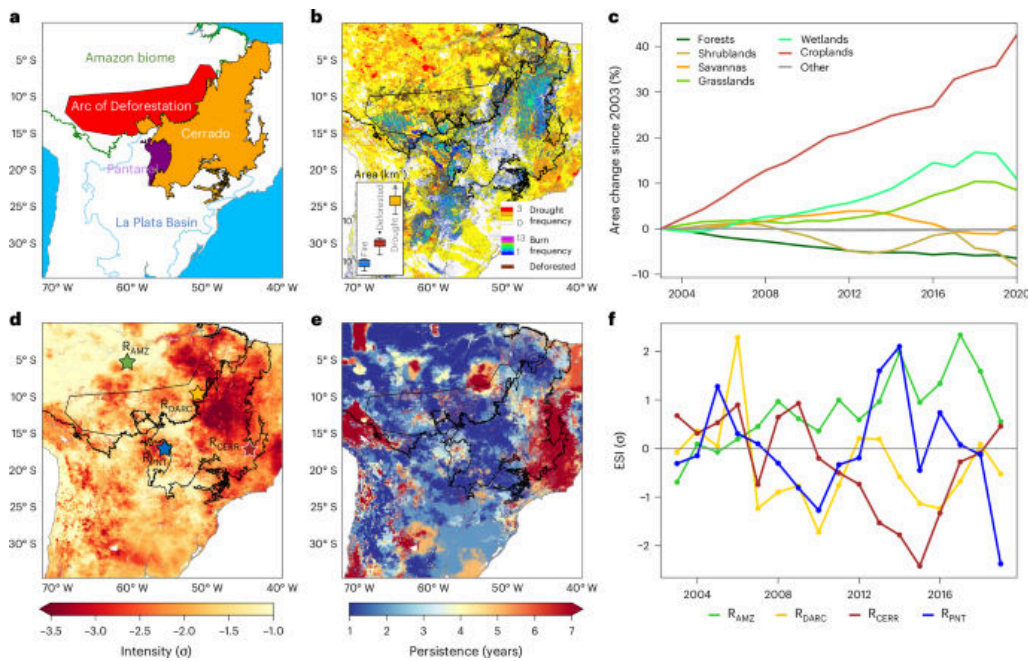
DOI: <https://link.springer.com/article/10.1007/s13593-026-01096-9>

环境与气候

5. Aligning climate change mitigation strategies with policy objectives beyond cost savings

(超越成本最优：让气候减缓路径与多重政策目标对齐)

Nature Climate Change | 2026-04-15 | 【Latest】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Climate Change, Published online: 15 April 2026; doi:10.1038/s41558-026-02617-w Optimal climate change mitigation pathways have historically focused on achieving emissions reductions while ensuring cost efficiency. However, the broader impacts of climate action are also important for policymakers and stakeholders. We developed a method that enables mitigation pathways to be defined based on their impact on multiple United Nations Sustainable Development Goals (SDGs).

中文摘要

研究提出一种新的路径设计方法，不再只以减排成本最低为目标，而是把多个联合国可持续发展目标（SDGs）的影响纳入评估。结果显示，兼顾发展、公平与环境协同效益，可以得到与传统成本最优路径显著不同的减排策略组合。

深度解读

传统减排路径常把‘最低成本’当成默认最优，但现实政策从来不只有一个目标。本文的重要价值在于把减排与粮食、安全、健康、公平等 SDG 指标放在同一张表里重新算账。结果说明，单纯追求最便宜的减排路径，可能把生态和社会代价转移到别的部门或群体。对中国和其他发展中经济体来说，这种多目标分配框架比单一成本最优更接近真实政策场景。

DOI: <https://doi.org/10.1038/s41558-026-02617-w>

土壤与生物地球化学

6. Dynamics of island mass effect –Part 2: Phytoplankton physiological responses

(岛屿质量效应动态之二：浮游植物的生理响应)

Biogeosciences | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Dynamics of island mass effect – Part 2: Phytoplankton physiological responses Guillaume Bourdin, Lee Karp-Boss, Fabien Lombard, Gabriel Gorsky, and Emmanuel Boss *Biogeosciences*, 23, 2687 – 2728, <https://doi.org/10.5194/bg-23-2687-2026>, 2026 Island mass effect (IME) refers to elevated chlorophyll a concentrations around islands, often extending hundreds of kilometers into oligotrophic waters. Using satellite-derived nutrient and iron stress indices and in situ bio-optical data from the Tara Pacific expedition, we show that IMEs are associated with recurring iron and macronutrient enrichments, and changes in phytoplankton community composition relative to the regional background ocean.

中文摘要

研究围绕“岛屿质量效应动态之二：浮游植物的生理响应”展开。英文摘要要点为：Dynamics of island mass effect –Part 2: Phytoplankton physiological responses Guillaume Bourdin, Lee Karp-Boss, Fabien Lombard, Gabriel Gorsky, and Emmanuel Boss *Biogeosciences*, 23, 2687–2728, <https://doi.org/10.5194/bg-23-2687-2026>, 2026 Island mass effect (IME) refers to elevated chlorophyll a concentrations around islands, often extending hundreds of kilometers into oligotrophic waters. Using satellite-derived nutrient and iron stress indices and in situ bio-optical data from the Tara Pacific expedition, we show that IMEs are associated with recurring iron and macronutrient enrichments, and changes in phytoplankton community composition relative to the regional background ocean. 该条目发表于 *Biogeosciences*，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.5194/bg-23-2687-2026>

环境与气候

7. More eddying of subtropical western boundary currents boosts stratification and cools shelf seas
(副热带西边界流涡旋增强会加剧层化并冷却陆架海)

Nature Climate Change | 2026-04-15 | **【Latest】**



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Climate Change, Published online: 15 April 2026; doi:10.1038/s41558-026-02599-9 This work shows that increased eddies accelerate surface warming in the Agulhas Current while also boosting hidden upwelling that cools the current, and adjacent shelf seas, at depth. Similar trends are expected for all subtropical western boundary currents, even if volume transports remain steady.

中文摘要

研究显示, 副热带西边界流中更频繁或更强的中尺度涡旋, 一方面会加速表层变暖, 另一方面也会增强隐蔽上升流, 使主流及邻近陆架海在深层出现冷却并加强水体层化。这种现象预计不只发生在阿古拉斯洋流, 也可能是类似洋流系统的普遍趋势。

深度解读

这项工作很有意思, 因为它揭示了海洋变暖并不总是表现为‘整层一起变热’。涡旋增强会把表层增温、深层冷却和层化加强同时推高, 导致近岸陆架海的生态环境更复杂。对渔业、营养盐输送和沿海低氧风险来说, 这种垂向结构变化往往比平均温度本身更关键。它提醒海洋管理不能只盯海表温度, 必须把中尺度过程和近岸生态响应一起看。

DOI: <https://doi.org/10.1038/s41558-026-02599-9>

保护与生物多样性

8. Distribution of the baraúna tree is threatened by climate change, land-cover change, and biological invasions in the Brazilian Caatinga

(气候变化、土地覆被变化与生物入侵共同威胁巴西旱林珍稀树种 baraúna)

Biodiversity and Conservation | 2026-04-16 | **【Latest】**

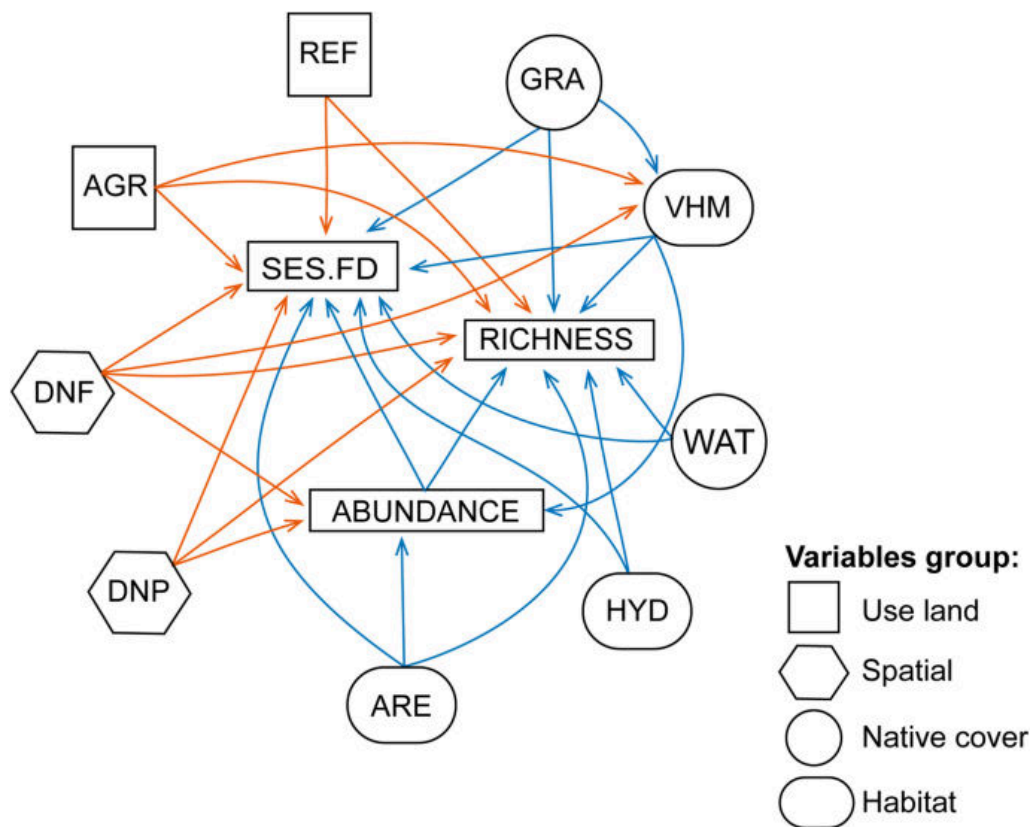


图 : Graphical abstract / article figure source: <https://static-content.springer.com/image/art%3A10.1007%2Fs10531-026-03340-v>

Abstract

Multiple drivers shape biodiversity loss worldwide. Land-use change and biological invasions are major direct pressures on species, while climate change increasingly alters species distributions and community dynamics, often intensifying other stressors. However, these factors are still insufficiently integrated into Ecological Niche Modeling (ENM) studies. We quantified the potentially suitable area for *Schinopsis brasiliensis* Engl., a tree species of high ecological importance in the Caatinga domain, considering the combined effects of climate change, land-cover change (LCC), and invasion by the exotic species *Neltuma juliflora* (Sw.) Raf. (hereafter, *Neltuma* spp.), known to reduce native plant abundance. We modeled the potential distributions of *S. brasiliensis* and *Neltuma* spp. under cur

中文摘要

研究将气候变化、土地覆被变化及生物入侵同时纳入生态位模型，评估巴西卡廷加地区关键树种 *Schinopsis brasiliensis* 的潜在适生区变化。结果显示，多重压力叠加将明显压缩其未来分布范围，并提高保护与恢复难度。

深度解读

这篇文章的重要之处在于，它没有把威胁因素拆开单独讨论，而是把气候、土地利用和入侵种放在同一个风险框架里。现实中的物种衰退往往就是这样发生的：不是单一因子致命，而是多重压力联手挤压。对于干旱区树种保护而言，这意味着保护工作不能只做保育地划定，还要同步考虑土地管理和入侵防控。文章也说明，未来的分布模型如果仍然只看气候，往往会低估真正的灭失风险。

DOI: <https://link.springer.com/article/10.1007/s10531-026-03340-w>

植物与农业

9. Climate change and crop-feeding herbivores

(气候变化与取食作物的植食性动物风险)

Nature Food | 2026-04-17 | 【Latest】

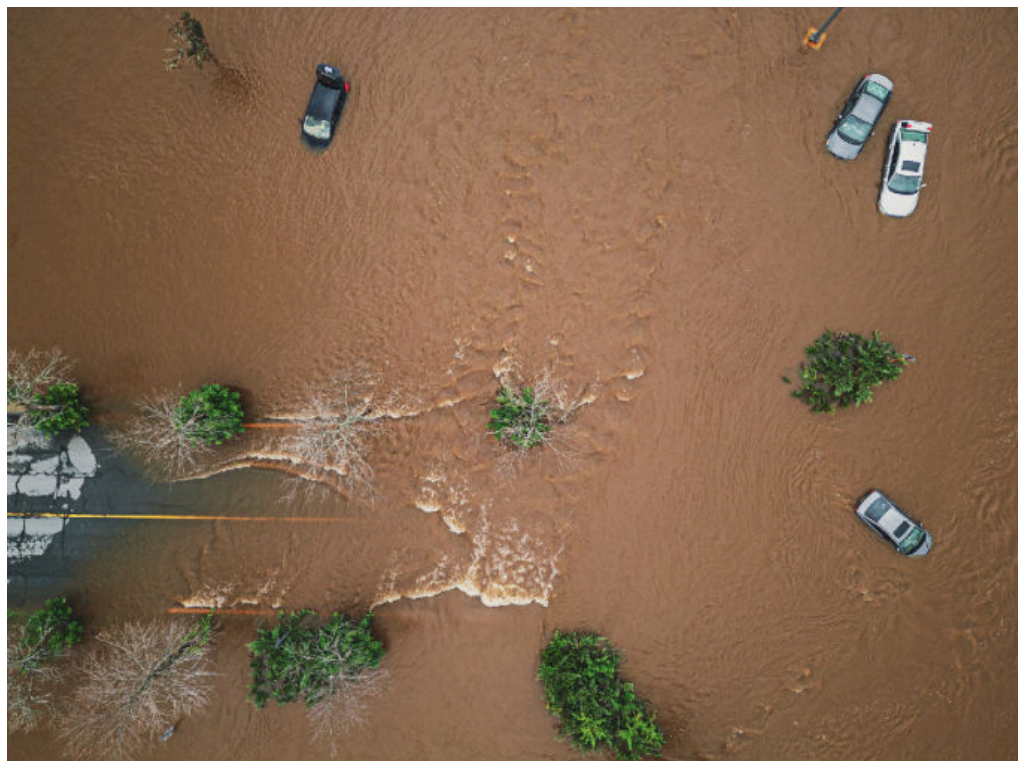


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Food, Published online: 17 April 2026; doi:10.1038/s43016-026-01346-9Climate change and crop-feeding herbivores

中文摘要

研究围绕“气候变化与取食作物的植食性动物风险”展开。英文摘要要点为: Climate change and crop-feeding herbivores 该条目发表于 Nature Food, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫, 对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s43016-026-01346-9>

环境与气候

10. Humanitarian blind spots in Western climate change policy and discourse

(西方气候政策与话语中的人道主义盲区)

Nature Climate Change | 2026-04-16 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Nature Climate Change, Published online: 16 April 2026; doi:10.1038/s41558-026-02613-0 Humanitarian blind spots in Western climate change policy and discourse

中文摘要

研究围绕“西方气候政策与话语中的人道主义盲区”展开。英文摘要要点为：Humanitarian blind spots in Western climate change policy and discourse 该条目发表于 Nature Climate Change，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41558-026-02613-0>

环境与气候

11. Turning the tide of climate solutions to save tidal flats

(扭转气候解决方案方向以拯救潮滩)

Nature Sustainability | 2026-04-20 | **【Latest】**



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 20 April 2026; doi:10.1038/s41893-026-01802-1 Tidal flats are vital carbon sinks and biodiversity hotspots, yet they face increasing threats from misapplied climate solutions, including mangrove afforestation and renewable energy expansion. Protecting tidal flats from well-intentioned but misapplied solutions is crucial for aligning global climate and biodiversity goals.

中文摘要

研究围绕“扭转气候解决方案方向以拯救潮滩”展开。英文摘要要点为：Tidal flats are vital carbon sinks and biodiversity hotspots, yet they face increasing threats from misapplied climate solutions, including mangrove afforestation and renewable energy expansion. Protecting tidal flats from well-intentioned but misapplied solutions is crucial for aligning global climate and biodiversity goals. 该条目发表于 Nature Sustainability，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41893-026-01802-1>

植物与农业

12. Higher optimal temperature for vegetation transpiration than for photosynthesis

(植被蒸腾的最适温度高于光合作用的最适温度)

Nature Plants | 2026-04-15 | **【Latest】**

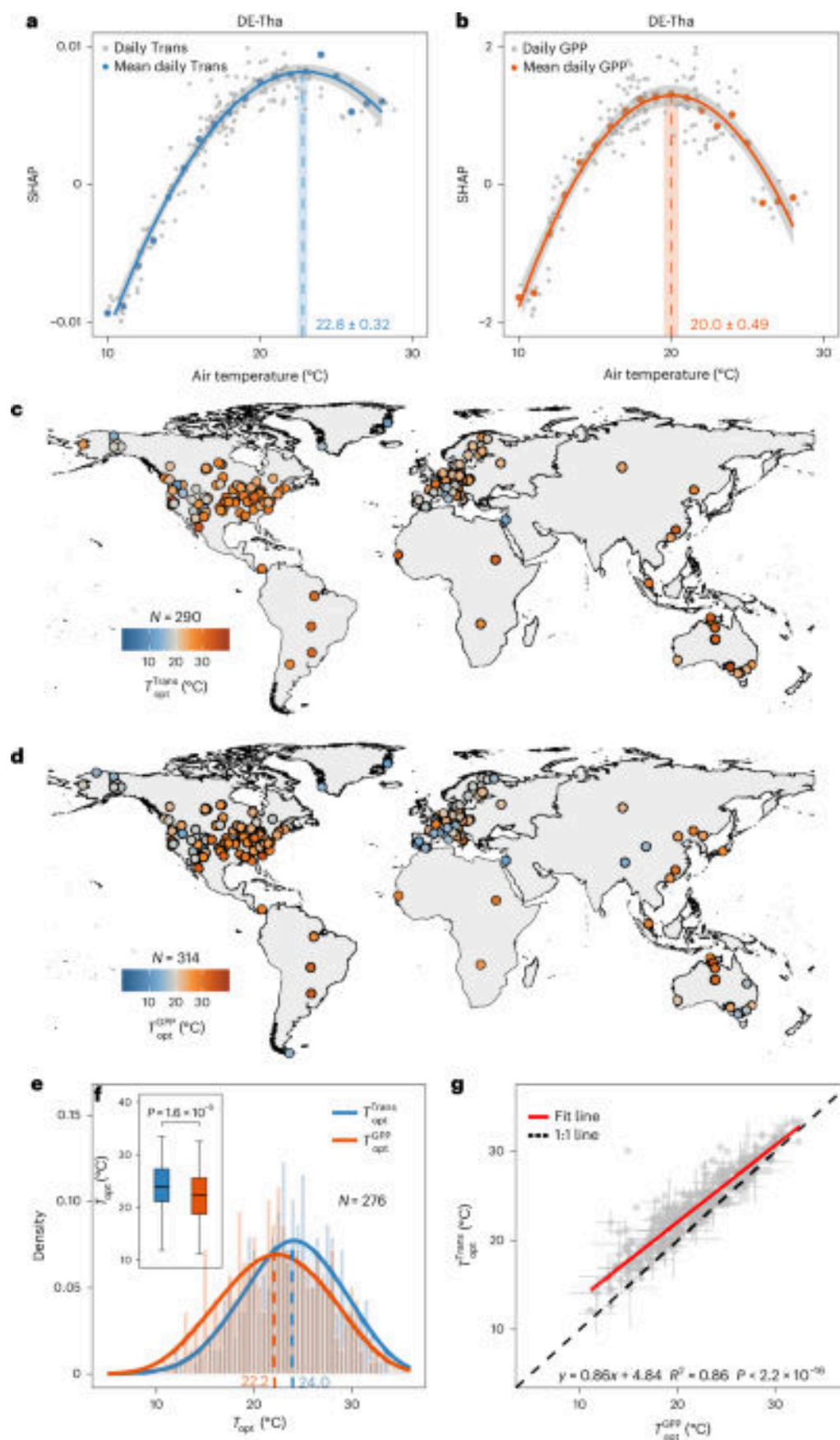


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Plants, Published online: 15 April 2026; doi:10.1038/s41477-026-02263-2 Global observations show that ecosystems sustain transpiration at temperatures above those that maximize photosynthesis. This thermal gap reveals that carbon uptake declines before water loss under climate warming.

中文摘要

全球观测表明，生态系统维持蒸腾作用的最适温度普遍高于光合作用达到峰值的温度。这意味着在持续变暖背景下，碳吸收会先于水分散失开始下降，暴露出生态系统‘先失碳、后失水’的热响应差异。

深度解读

这篇 Nature Plants 文章抓住了陆地生态系统对变暖响应中的一个关键不对称性：植物在高温下会先失去固碳效率，却还在继续耗水。换句话说，生态系统可能更早从‘高效碳汇’变成‘低效用水者’。这对于干旱风险评估、作物模型和全球碳循环预测都非常重要，因为很多模型默认碳—水响应是同步的。若这种热阈值错位普遍存在，未来高温对生态系统功能的冲击可能比预想更早到来。

DOI: <https://doi.org/10.1038/s41477-026-02263-2>

综合顶刊

13. Leveraging remote sensing and crowd-sourced biodiversity data for enhanced plant functional trait mapping

(融合遥感与众包生物多样性数据可提升全球植物功能性状制图)

Nature Communications | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Nature Communications, Published online: 21 April 2026; doi:10.1038/s41467-026-72111-6 Plant functional traits govern ecosystem processes but lack spatially detailed global representations. Here, the authors combine remote sensing and crowd-sourced biodiversity data to map leaf trait distributions and their higher-order moments worldwide.

中文摘要

研究结合遥感观测与众包生物多样性数据，绘制了全球尺度叶片功能性状及其高阶统计矩分布图。该方法显著提升了植物功能性状的空间表达能力，为理解生态系统功能和全球变化响应提供了更精细的数据基础。

深度解读

植物功能性状一直是连接物种组成与生态系统过程的关键桥梁，但过去最大的问题是空间分辨率不够。本文把遥感和大规模众包生物多样性数据结合起来，相当于给全球植物性状地图补上了‘细节’。

层’。这不仅有助于提高碳循环、植被动态和生态系统服务模型的真实性和完整性，也说明公民科学数据在全球生态监测中的价值越来越高。对未来生态预测来说，‘有性状但没地图’这个老短板正在被补上。

DOI: <https://doi.org/10.1038/s41467-026-72111-6>

土壤与生物地球化学

14. Potential of optical and ecological proxies to quantify phytoplankton carbon in oligotrophic waters

(利用光学与生态代理指标量化贫营养水体浮游植物碳)

Biogeosciences | 2026-04-20 | **【Latest】**



图：Graphical abstract / article figure source:

Abstract

Potential of optical and ecological proxies to quantify phytoplankton carbon in oligotrophic waters David Antoine, Chandanlal Parida, and Camille Grimaldi Biogeosciences, 23, 2641 – 2660, <https://doi.org/10.5194/bg-23-2641-2026>, 2026 A dataset of phytoplankton cell counts, pigments, particulate organic carbon and optical properties enables comparison of three methods to estimate phytoplankton carbon (C_{phyto}) in oligotrophic waters, where uncertainties in phytoplankton productivity are still large. Two methods based on chlorophyll concentration and particulate backscattering, are scalable to global scale while cell counts reduce bias from non-algal material. This comparison clarifies uncertainties in optical C_{phyto} estimates.

中文摘要

研究围绕“利用光学与生态代理指标量化贫营养水体浮游植物碳”展开。英文摘要要点为：Potential of optical and ecological proxies to quantify phytoplankton carbon in oligotrophic waters David Antoine, Chandanlal Parida, and Camille Grimaldi Biogeosciences, 23, 2641 – 2660, <https://doi.org/10.5194/bg-23-2641-2026>, 2026 A dataset of phytoplankton cell counts, pigments, particulate organic carbon and optical properties enables comparison of three methods to estimate phytoplankton carbon (C_{phyto}) in oligotrophic waters, where uncertainties in phytoplankton productivity are still large. Two methods based on chlorophyll concentration and particulate backscattering, are scalable to global scale while cell counts reduce bias from non-algal material. This comparison clarifies uncertainties in optical C_{phyto} estimates. 该条目发表于 Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

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DOI: <https://doi.org/10.5194/bg-23-2641-2026>

土壤与生物地球化学

15. Soil moisture-induced changes in land carbon sink projections in CMIP6

(土壤水分过程会显著改变 CMIP6 对陆地碳汇的未来预测)

Biogeosciences | 2026-04-21 | 【Latest】



图： Graphical abstract / article figure source:

Abstract

Soil moisture-induced changes in land carbon sink projections in CMIP6 Lea Maria Gabele, Petra Sieber, Laibao Liu, and Sonia I. Seneviratne Biogeosciences, 23, 2729–2746, <https://doi.org/10.5194/bg-23-2729-2026>, 2026 We investigate how soil moisture influences the future land carbon sink using dedicated experiments from the latest generation of Earth system models. Our results show that soil moisture strongly reduces land carbon uptake, mainly through soil moisture–atmosphere coupling. However, models disagree on the magnitude of SM impacts. The study emphasises that improving the representation of plant water stress in models is essential to reduce uncertainties in carbon cycle projections.

中文摘要

研究利用新一代地球系统模型专门试验，评估土壤水分如何影响未来陆地碳汇。结果显示，土壤水分主要通过土壤—大气耦合显著抑制陆地碳吸收，但不同模型对影响强度差异很大，说明植物水分胁迫表征仍是碳循环预测中的关键不确定源。

深度解读

陆地碳汇预测常常把注意力放在二氧化碳浓度和温度上，但这篇文章提醒我们，土壤水分可能是更硬的限制条件。它不仅影响植物生长，还会通过陆气耦合放大或重塑整个碳吸收过程。更麻烦的是，不同模型在这一点上分歧很大，说明目前的未来碳汇判断还带着明显的结构性不确定性。对气候目标评估而言，这意味着我们不能对陆地持续吸碳过于乐观。

DOI: <https://doi.org/10.5194/bg-23-2729-2026>

土壤与生物地球化学

16. Technical note: Development of an extraction protocol and colorimetric analysis for alginate in marine sediment

(技术札记：海洋沉积物中海藻酸盐提取与比色分析方法)

Biogeosciences | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Technical note: Development of an extraction protocol and colorimetric analysis for alginate in marine sediment Sota Nakazato, Takashi Kawahara, Reiji Ichii, Shingo Akita, Akira Inoue, Masaki Fujita, Hideyuki Kurihara, Mitsuhiro Nakaya, and Atsushi Ooki Biogeosciences, 23, 2747–2759, <https://doi.org/10.5194/bg-23-2747-2026>, 2026 Seaweeds can store carbon in the ocean, but traditional methods in food chemistry fail to measure alginate—a major organic component derived from brown algae—in seafloor mud. We developed a novel extraction method for alginate in marine sediments. This allows for high recovery rates, and we successfully used this method to quantify alginate in marine sediments from northern Japan. Our work establishes a reliable method to track alginate on the ocean floor.

中文摘要

研究围绕“技术札记：海洋沉积物中海藻酸盐提取与比色分析方法”展开。英文摘要要点为：Technical note: Development of an extraction protocol and colorimetric analysis for alginate in marine sediment Sota Nakazato, Takashi Kawahara, Reiji Ichii, Shingo Akita, Akira Inoue, Masaki Fujita, Hideyuki Kurihara, Mitsuhiro Nakaya, and Atsushi Ooki Biogeosciences, 23, 2747–2759, <https://doi.org/10.5194/bg-23-2747-2026>, 2026 Seaweeds can store carbon in the ocean, but traditional methods in food chemistry fail to measure alginate—a major organic component derived from brown algae—in seafloor mud. We developed a novel extraction method for alginate in marine sediments. This allows for high recovery rates, and we successfully used this method to quantify alginate in marine sediments from northern Japan. Our work establishes a reliable method to track alginate on the ocean floor. 该条目发表于 Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.5194/bg-23-2747-2026>

环境与气候

17. Decades of increased emissions from forest-fuelled BECCS

(以现有森林木材为燃料的 BECCS 可能在数十年内反而增加排放)

Nature Sustainability | 2026-04-20 | **【Latest】**

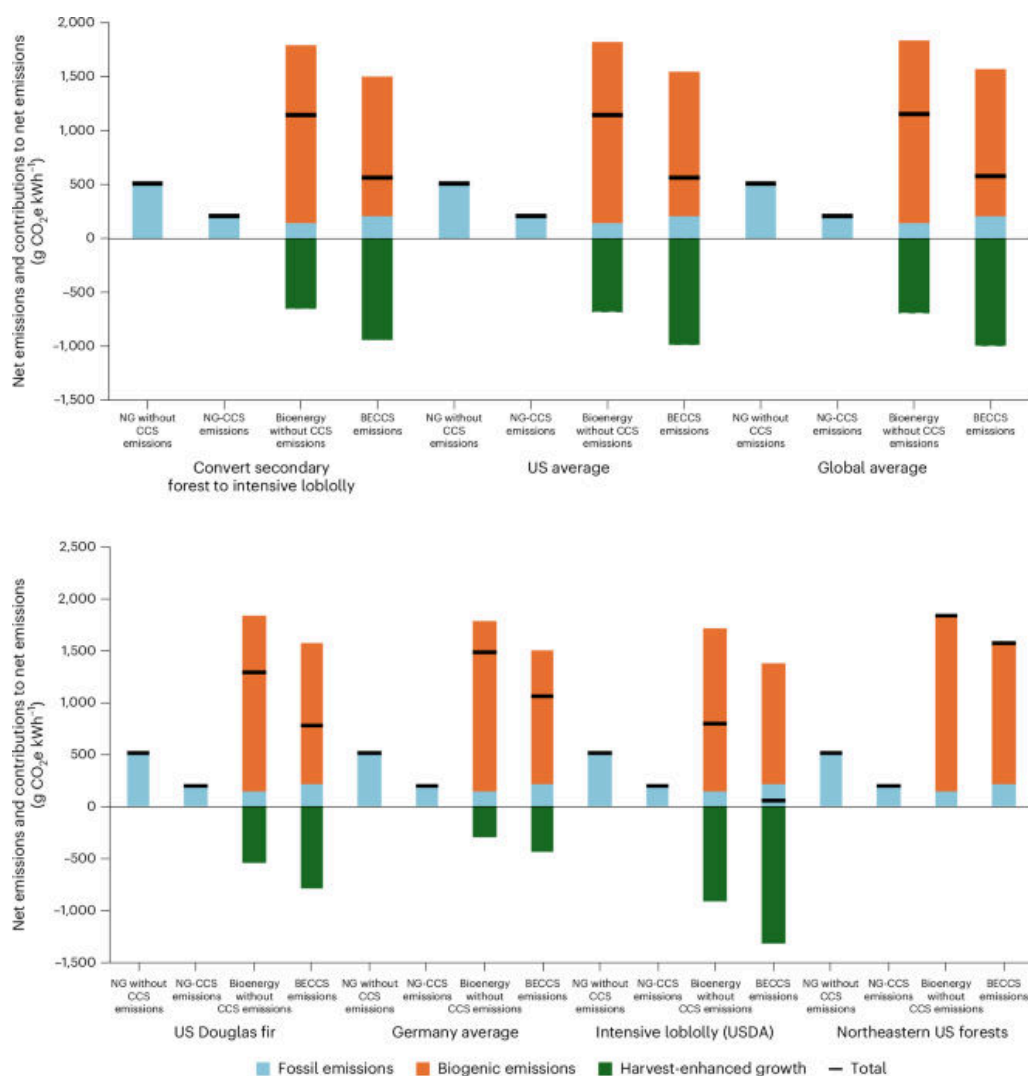


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 20 April 2026; doi:10.1038/s41893-026-01817-8 Governments are considering subsidies for bioenergy with carbon capture and storage fuelled by wood from existing forests. However, a transparent model estimates that this will probably increase emissions for decades, increase electricity costs 3.5-fold and not generate negative emissions for 150 years.

中文摘要

研究用透明模型评估以现有森林木材供能的生物质能碳捕集与封存（BECCS）方案，发现该路径可能在数十年内提高净排放、电力成本增加约 3.5 倍，并且约 150 年后才可能出现负排放效果。

深度解读

这篇文章直指负排放讨论中的一个痛点：并不是所有 BECCS 都天然减碳。若燃料来自现有森林，账面上的‘可再生’很可能掩盖了长期碳债，结果是前几十年排放更高、成本更贵、气候收益却极晚才出现。它的重要意义在于，把时间尺度重新拉回政策视野——对 2030 或 2050 目标来说，150 年后的收益几乎没有决策价值。对生物质补贴和碳中和路径设计，这是一记很实在的提醒。

DOI: <https://doi.org/10.1038/s41893-026-01817-8>

遥感与模型

18. Monitoring Coral Reef Metabolism Under Changing Oceans – Novel Insights From Seawater Stable Carbon Isotopes

(利用海水稳定碳同位素监测变化海洋中的珊瑚礁代谢)

Journal of Geophysical Research - Biogeosciences | 2026-04-18 | **【Latest】**



图： Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“利用海水稳定碳同位素监测变化海洋中的珊瑚礁代谢”展开。英文摘要要点为： Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于遥感与模型方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009416>

海洋与水生

19. Energy-mediated feedbacks of vegetation greening enhance precipitation efficiency and sustain water yield in global semi-arid regions

(植被变绿通过能量反馈提高降水效率并维持全球半干旱区水量产出)

Nature Water | 2026-04-17 | **【Latest】**

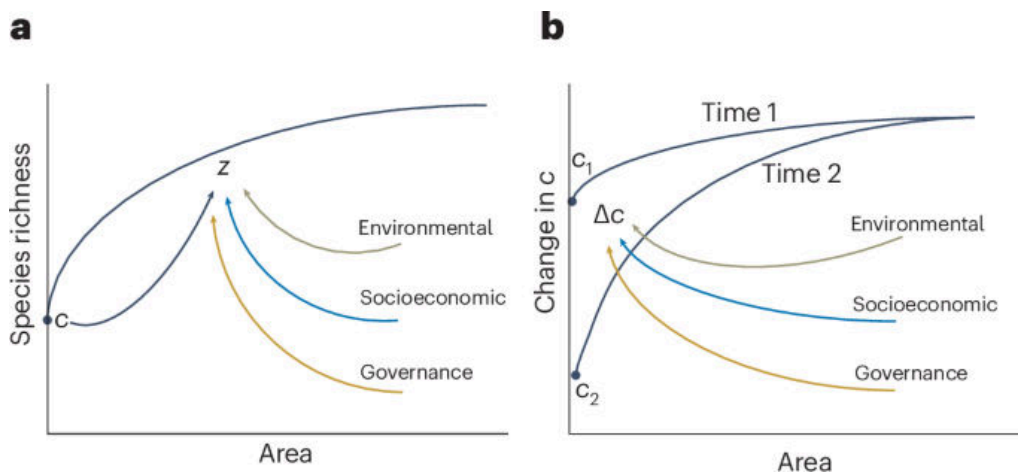


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Water, Published online: 17 April 2026; doi:10.1038/s44221-026-00631-yVegetation greening is traditionally thought to increase evapotranspiration and reduce streamflow, posing a challenge for balancing ecosystem restoration with water supply. This study reveals that greening can enhance both evapotranspiration and streamflow, particularly in semi-arid regions through albedo-driven, energy-mediated increases in the efficiency with which atmospheric moisture is converted into precipitation.

中文摘要

传统观点认为植被恢复会因蒸散增强而减少径流，但该研究发现，在全球半干旱区，植被变绿可通过反照率变化等能量反馈提高大气水汽转化为降水的效率，从而同时提升蒸散和水量产出。

深度解读

这项研究之所以重要，是因为它挑战了‘绿了就更耗水’的单向叙事。文章表明，在半干旱地区，植被恢复并不一定与水资源目标冲突，关键在于能量收支和区域降水反馈如何被重新组织。换句话说，生态修复可能不是简单地从河流‘拿走水’，而是在某些区域反过来提升水循环效率。对退化治理、三北工程和干旱区流域管理来说，这类结果很值得深入跟进。

DOI: <https://doi.org/10.1038/s44221-026-00631-y>

保护与生物多样性

20. From species to therapeutic property: distribution differences of threatened Chinese medicinal plants under climate change

(在气候变化下，受威胁中国药用植物的分布格局及其药效类别差异正在重组)

Biodiversity and Conservation | 2026-04-16 | 【Latest】

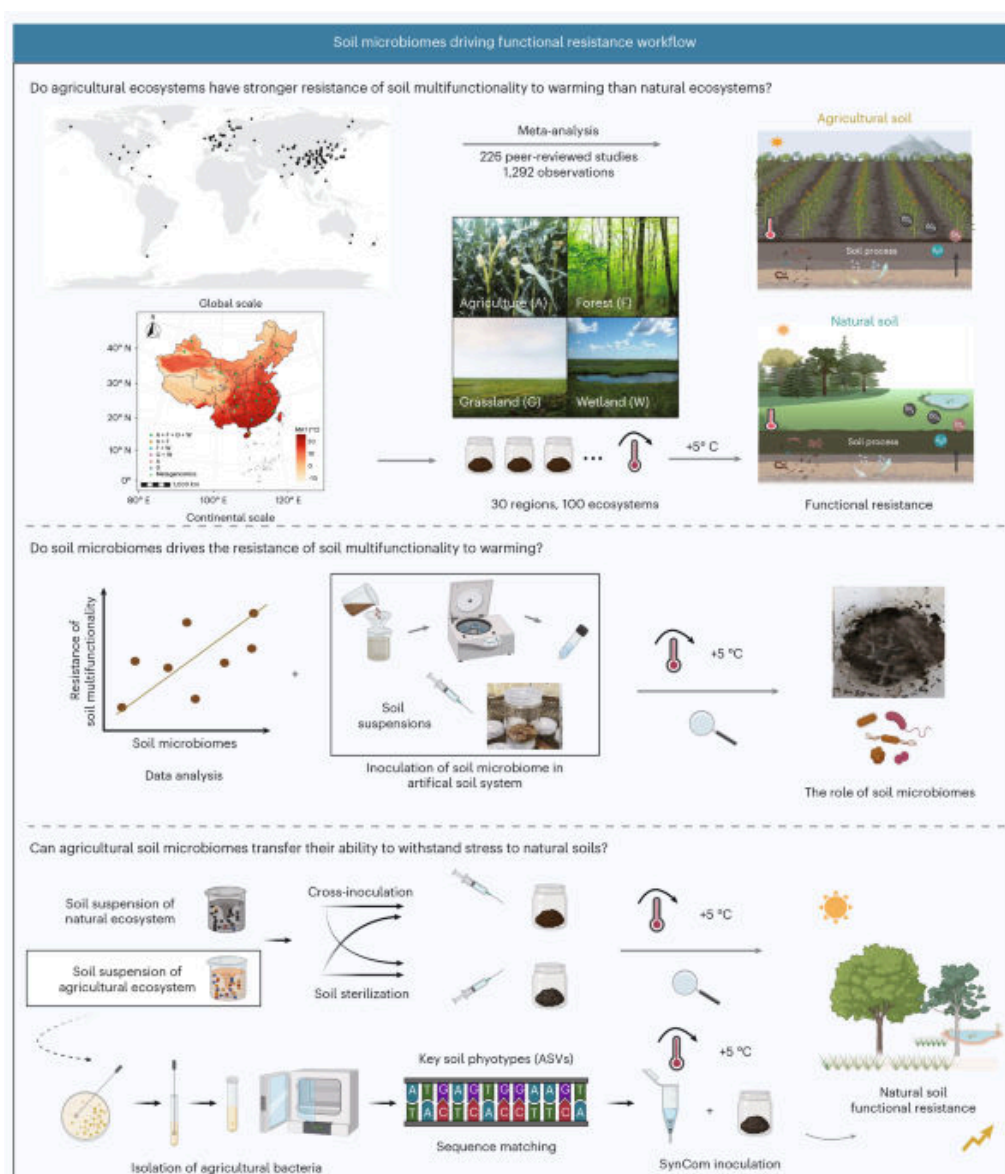


图 : Graphical abstract / article figure source: <https://static-content.springer.com/image/art%3A10.1007%2Fs10531-026-03350-8>

Abstract

Once medicinal plants become extinct, their unique therapeutic property will also be extinguished. At present, conservation strategies focus on overall species richness (SR), but ignore the impacts of climate change on different therapeutic categories. This study aims to explore the differences in responses of various therapeutic categories to climate change and provide scientific references for more targeted protection of medicinal resources. Using a list of 757 threatened Chinese medicinal plants (CMPs), species distribution ensemble models were used to simulate the distribution patterns of each species in the current and future (2081–2100). The analysis focused on SR and ecological niche overlap across eight therapeutic categories. It identified key environmental driving factors and del

中文摘要

研究以 757 种受威胁中国药用植物为对象，利用集成物种分布模型模拟当前与 2081–2100 年的分布变化，并比较八类治疗功效植物在物种丰富度和生态位重叠上的差异。结果表明，不同药效类别对气候变化的脆弱性并不一致，保护策略不能只看总体物种数。

深度解读

这篇文章把生物多样性保护和药用资源安全真正连到了一起。过去很多研究只问某个类群会不会减少，但这里进一步追问：如果丢失的是某类关键药效资源，社会和健康层面的代价会是什么。它说

明药用植物保护不能只按物种丰富度排序，而要把功能和用途维度一起考虑。对中药资源保育、迁地保护和种质库建设，这个思路非常实际。

DOI: <https://link.springer.com/article/10.1007/s10531-026-03350-8>

保护与生物多样性

21. Enhancement of climate-related plant translocations by reinforcing populations with nearby genotypes
(利用邻近基因型强化种群以提升气候相关植物迁地保护成效)

Conservation Biology | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“利用邻近基因型强化种群以提升气候相关植物迁地保护成效”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70289>

保护与生物多样性

22. Delayed plant genetic responses to forest edge dynamics and their conservation implications
(植物对森林边缘动态的遗传响应滞后及其保护启示)

Conservation Biology | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“植物对森林边缘动态的遗传响应滞后及其保护启示”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

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DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70297>

生态学核心

23. Multicellularity for free

(多细胞性演化的低成本路径)

Nature Ecology & Evolution | 2026-04-20 | **【Latest】**

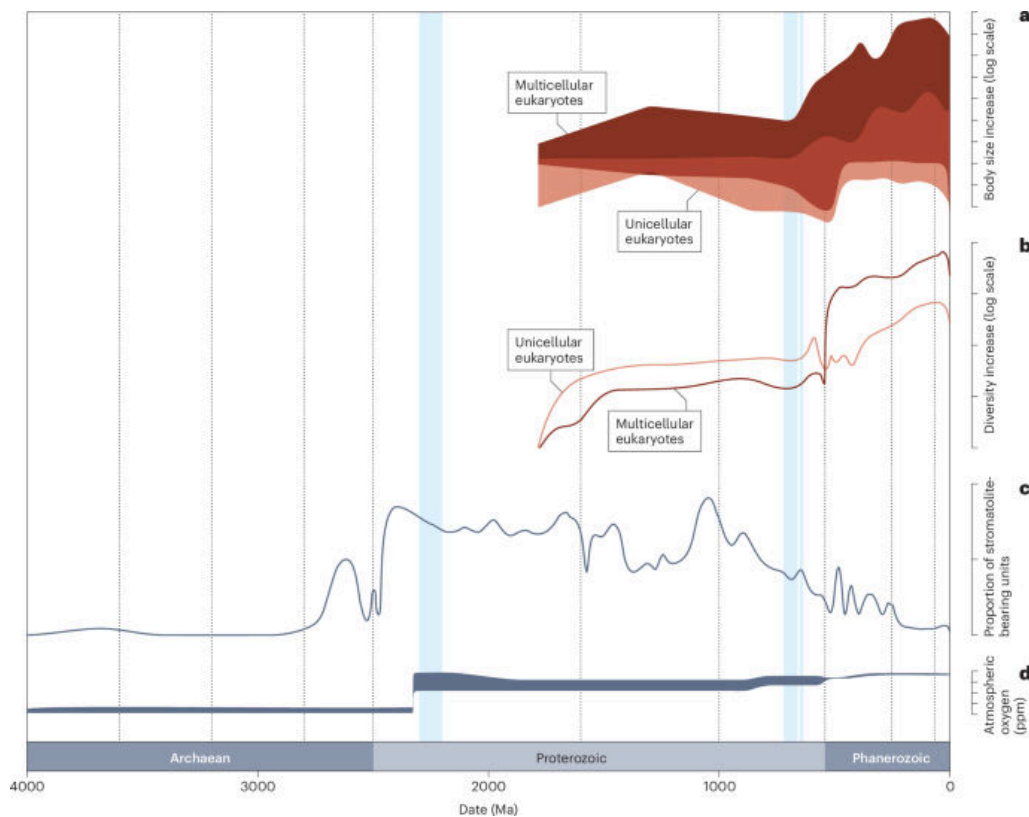


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Ecology & Evolution, Published online: 20 April 2026; doi:10.1038/s41559-026-03021-5A model of competition between unicellular and multicellular forms suggests that multicellularity can become ecologically stable without direct benefits, provided both forms use a varied environment in different ways.

中文摘要

研究围绕“多细胞性演化的低成本路径”展开。英文摘要要点为：A model of competition between unicellular and multicellular forms suggests that multicellularity can become ecologically stable without direct benefits, provided both forms use a varied environment in different ways. 该条目发表于 Nature Ecology & Evolution，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

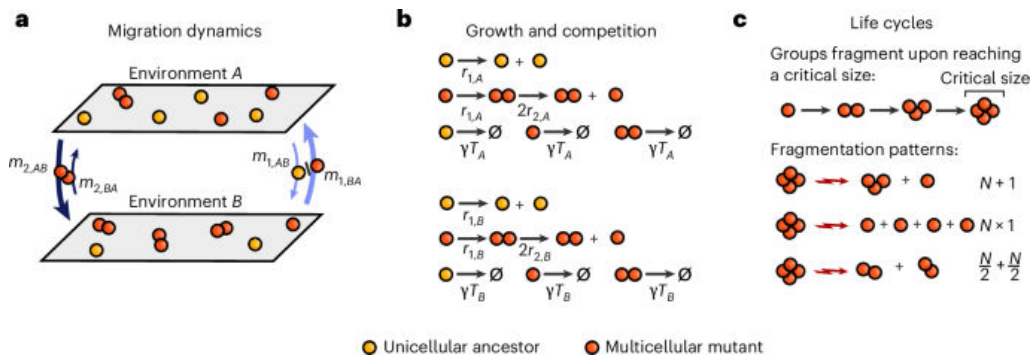
DOI: <https://doi.org/10.1038/s41559-026-03021-5>

生态学核心

24. Direct benefits are not necessary for the evolution of multicellularity

(多细胞性演化并不必然需要直接收益)

Nature Ecology & Evolution | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Ecology & Evolution, Published online: 20 April 2026; doi:10.1038/s41559-026-03044-y A model for the evolution of multicellularity in a spatially heterogeneous environment shows that indirect mechanisms such as escape from competition and increased environmental exploitation can drive the evolution of multicellularity in the absence of direct benefits.

中文摘要

研究围绕“多细胞性演化并不必然需要直接收益”展开。英文摘要要点为：A model for the evolution of multicellularity in a spatially heterogeneous environment shows that indirect mechanisms such as escape from competition and increased environmental exploitation can drive the evolution of multicellularity in the absence of direct benefits. 该条目发表于 Nature Ecology & Evolution，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41559-026-03044-y>

环境与气候

25. [ASAP] MarineTox Predictor: An Online Library Platform for Enhancing Low-Resourced Saltwater Ecotoxicity Prediction via Knowledge Sharing from Freshwater Ecotoxicity

(MarineTox Predictor：借助淡水毒性知识提升海水生态毒性预测的在线平台)

Environmental Science & Technology | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c15496

中文摘要

研究围绕“MarineTox Predictor：借助淡水毒性知识提升海水生态毒性预测的在线平台”展开。英文摘要要点为：Environmental Science & TechnologyDOI: 10.1021/acs.est.5c15496 该条目发表于 Environmental Science & Technology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

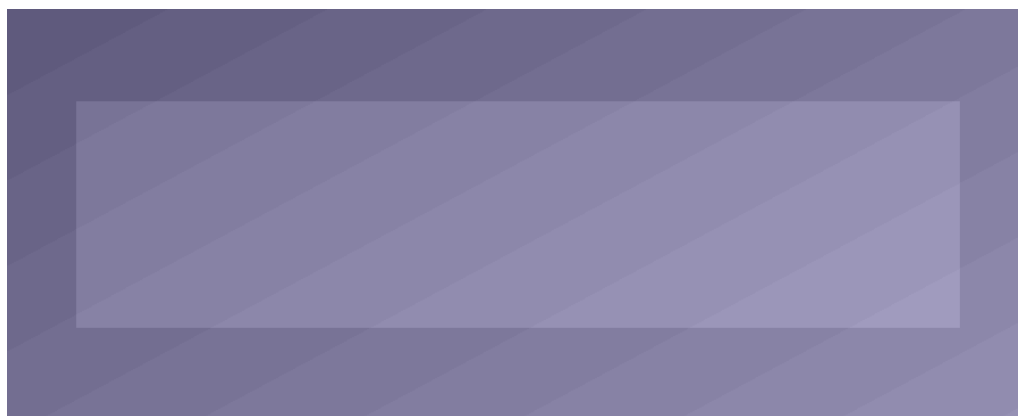
该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1021/acs.est.5c15496>

土壤与生物地球化学

26. Improving coastal ocean pH estimates through assimilation of glider observations and hybrid statistical methods
(融合滑翔机观测与混合统计方法改进近岸海洋 pH 估算)

Biogeosciences | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Improving coastal ocean pH estimates through assimilation of glider observations and hybrid statistical methods Jann Paul Mattern, Yuichiro Takeshita, Carlos Rocha, and Christopher A. Edwards Biogeosciences, 23, 2621–2639, <https://doi.org/10.5194/bg-23-2621-2026>, 2026 We improve coastal ocean carbonate system estimates by assimilating glider pH and alkalinity data into a regional biogeochemical model. Joint assimilation with physical observations successfully improves pH estimates while maintaining physical estimates. A hybrid approach combining dynamical models with statistical methods produces accurate pH estimates without requiring biogeochemical models, offering an alternative solution for ocean acidification monitoring.

中文摘要

研究围绕“融合滑翔机观测与混合统计方法改进近岸海洋 pH 估算”展开。英文摘要要点为：Improving coastal ocean pH estimates through assimilation of glider observations and hybrid statistical methods Jann Paul Mattern, Yuichiro Takeshita, Carlos Rocha, and Christopher A. Edwards Biogeosciences, 23, 2621–2639,

<https://doi.org/10.5194/bg-23-2621-2026>, 2026 We improve coastal ocean carbonate system estimates by assimilating glider pH and alkalinity data into a regional biogeochemical model. Joint assimilation with physical observations successfully improves pH estimates while maintaining physical estimates. A hybrid approach combining dynamical models with statistical methods produces accurate pH estimates without requiring biogeochemical models, offering an alternative solution for ocean acidification monitoring. 该条目发表于 Biogeosciences, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程, 有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.5194/bg-23-2621-2026>

保护与生物多样性

27. Relative importance of traits, climate, and threats to extinction risk in salamanders

(性状、气候与威胁因素对蝾螈灭绝风险的相对重要性)

Conservation Biology | 2026-04-19 | **【Latest】**



图: Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“性状、气候与威胁因素对蝾螈灭绝风险的相对重要性”展开。英文摘要要点为: Conservation Biology, EarlyView. 该条目发表于 Conservation Biology, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效, 强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70281>

生态学核心

28. Environmental Fluctuations Are Insufficient to Explain High Coral Biodiversity: A Test of Five Storage Effect Pathways

(环境波动不足以解释高珊瑚多样性：五种储存效应路径检验)

Ecology Letters | 2026-04-17 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“环境波动不足以解释高珊瑚多样性：五种储存效应路径检验”展开。英文摘要要点为：Environmental Fluctuations Are Insufficient to Explain High Coral Biodiversity: A Test of Five Storage Effect Pathways 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

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DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70362>

遥感与模型

29. Hydrological Mismatch in Arid Planted Shrublands: Non-Responsiveness to Precipitation Changes and Unsustainable Water Use

(干旱人工灌丛中的水文错配：对降水变化不敏感与不可持续用水)

Journal of Geophysical Research - Biogeosciences | 2026-04-17 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“干旱人工灌丛中的水文错配：对降水变化不敏感与不可持续用水”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

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DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2026JG009715>

生态学核心

30. Environmental Variability Shapes Life-History Trade-Offs Within and Between Populations of a Long-Lived Seabird

(环境变异塑造长寿海鸟种群内外的生活史权衡)

Ecology Letters | 2026-04-16 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“环境变异塑造长寿海鸟种群内外的生活史权衡”展开。英文摘要要点为：Environmental Variability Shapes Life-History Trade-Offs Within and Between Populations of a Long-Lived Seabird 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

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DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70384>

环境与气候

31. [ASAP] Accounting for Differences in Plasma Protein Binding between Fish and Humans for Supporting Environmental Risk Assessment of Pharmaceuticals Using the Fish Plasma Model

(在鱼血浆模型中纳入鱼类与人类血浆蛋白结合差异以支持药物环境风险评估)

Environmental Science & Technology | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Environmental Science & Technology DOI: 10.1021/acs.est.5c15513

中文摘要

研究围绕“在鱼血浆模型中纳入鱼类与人类血浆蛋白结合差异以支持药物环境风险评估”展开。英文摘要要点为：Environmental Science & Technology DOI: 10.1021/acs.est.5c15513 该条目发表于 Environmental Science & Technology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1021/acs.est.5c15513>

遥感与模型

32. Reducing Uncertainties in Net Carbon Capture to Advance Wetlands as Natural Climate Solutions (降低净碳捕获不确定性以推动湿地成为自然气候解决方案)

Journal of Geophysical Research - Biogeosciences | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“降低净碳捕获不确定性以推动湿地成为自然气候解决方案”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于遥感与模型方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009136>

土壤与生物地球化学

33. Organic Carbon and Texture Control Moisture Dependence of Soil Shortwave Infrared Reflectance (有机碳和质地控制土壤短波红外反射率的水分依赖性)

European Journal of Soil Science | 2026-04-19 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“有机碳和质地控制土壤短波红外反射率的水分依赖性”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science, 报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70319>

生态学核心

34. Advances in Modelling Radiative Transfer, Heat Storage and Turbulent Transport to Evaluate CO₂, Heat and Water Fluxes Over Broad-Leaved Forests: The CanVeg2 Model

(CanVeg2 模型改进阔叶林二氧化碳、热量与水分通量模拟)

Global Change Biology | 2026-04-18 | **【Latest】**



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“CanVeg2 模型改进阔叶林二氧化碳、热量与水分通量模拟”展开。英文摘要要点为: Advances in Modelling Radiative Transfer, Heat Storage and Turbulent Transport to Evaluate CO₂, Heat and Water Fluxes Over Broad-Leaved Forests: The CanVeg2 Model 该条目发表于 Global Change Biology, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程, 提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70867>

综合顶刊

35. Historical and contemporary genomes of an endangered rodent reveal shifts in environmentally associated genes

(濒危啮齿动物历史与现代基因组揭示环境关联基因变化)

Science Advances | 2026-04-17 | **【Latest】**



图: Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“濒危啮齿动物历史与现代基因组揭示环境关联基因变化”展开。英文摘要要点为: Historical and contemporary genomes of an endangered rodent reveal shifts in environmentally associated genes 该条目发表于 Science Advances, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程, 提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于综合顶刊方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1126/sciadv.adz5059>

海洋与水生

36. A renewable tango for drought-stricken power grids (干旱冲击下电网与可再生能源的协同适应)

Nature Water | 2026-04-16 | 【Latest】

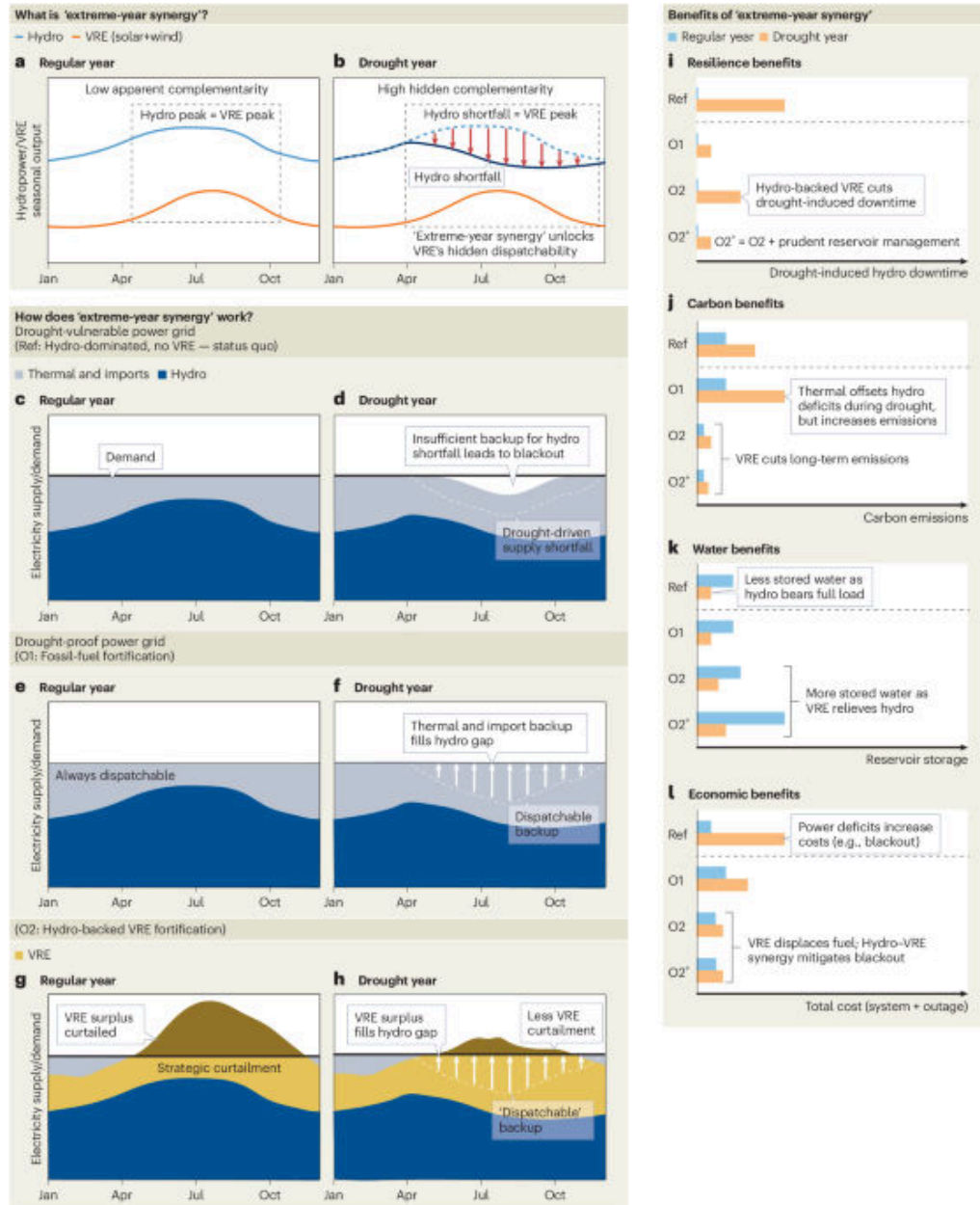


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Water, Published online: 16 April 2026; doi:10.1038/s44221-026-00625-w Like a tango partner stepping forward when the other falters, solar and wind can compensate for drought-driven hydropower shortfalls —turning these variable renewables into an underappreciated source of grid resilience.

中文摘要

研究围绕“干旱冲击下电网与可再生能源的协同适应”展开。英文摘要要点为：Like a tango partner stepping forward when the other falters, solar and wind can compensate for drought-driven hydropower shortfalls

—turning these variable renewables into an underappreciated source of grid resilience. 该条目发表于 Nature Water, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置, 对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s44221-026-00625-w>

土壤与生物地球化学

37. Inorganic Carbon (Calcite and Dolomite) Determination Using the Modified Pressure-Calcimeter Method
(基于改进压力钙量计法测定方解石和白云石无机碳)

European Journal of Soil Science | 2026-04-16 | 【Latest】



图: Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“基于改进压力钙量计法测定方解石和白云石无机碳”展开。英文摘要要点为: European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程, 有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70332>

环境与气候

38. The linkage between microbial community dynamics and urbanization age (微生物群落动态与城市化年限之间的联系)

Nature Sustainability | 2026-04-15 | 【Latest】

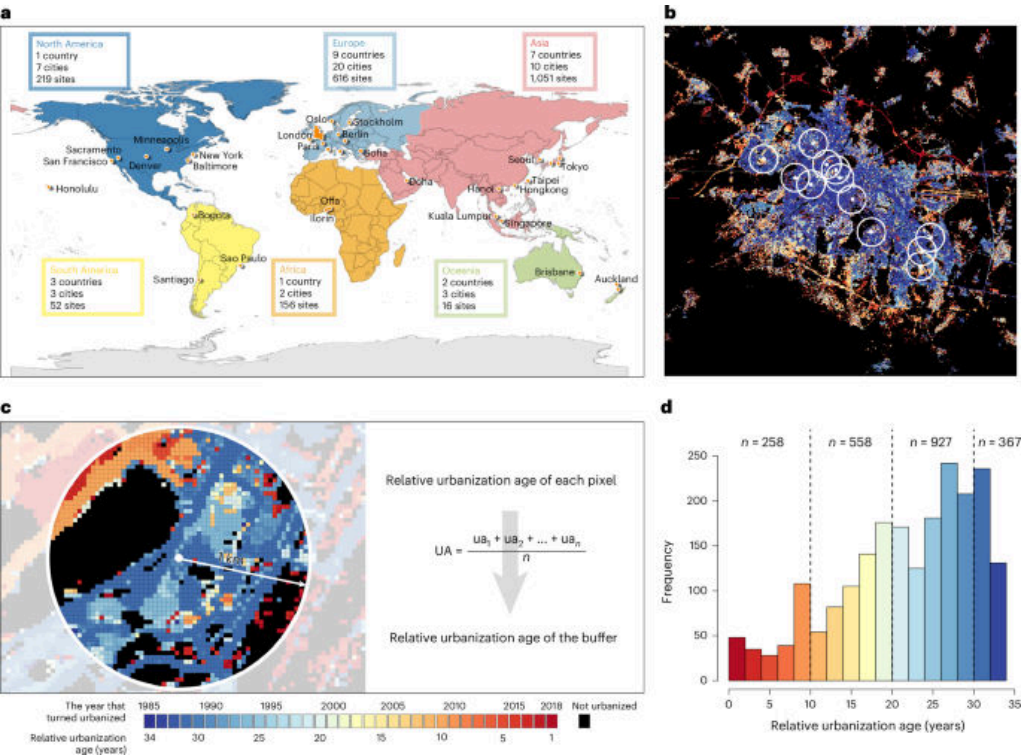


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Sustainability, Published online: 15 April 2026; doi:10.1038/s41893-026-01801-2 Urbanization can affect the dynamics of microbial communities, which can, in turn, have implications for public health. By leveraging global metagenome datasets, the authors study the impact of urbanization age on microbial communities for actionable insights for urban planning and public health.

中文摘要

研究围绕“微生物群落动态与城市化年限之间的联系”展开。英文摘要要点为：Urbanization can affect the dynamics of microbial communities, which can, in turn, have implications for public health. By leveraging global metagenome datasets, the authors study the impact of urbanization age on microbial communities for actionable insights for urban planning and public health. 该条目发表于 Nature Sustainability，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41893-026-01801-2>

植物与农业

39. A dual sustainability crisis for Kenya's small-scale fisheries
(肯尼亚小规模渔业面临双重可持续性危机)

Nature Food | 2026-04-14 | 【Latest】



图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Food, Published online: 14 April 2026; doi:10.1038/s43016-026-01342-z Data collected over 27 years show that ecological and social sustainability can quietly diverge, driven by factors that fisheries assessments typically overlook. For small-scale reef fishers in Kenya, rising fish prices and higher catches have failed to translate into decent earnings, with fewer than 2% making the national minimum wage.

中文摘要

研究围绕“肯尼亚小规模渔业面临双重可持续性危机”展开。英文摘要要点为：Data collected over 27 years show that ecological and social sustainability can quietly diverge, driven by factors that fisheries assessments typically overlook. For small-scale reef fishers in Kenya, rising fish prices and higher catches have failed to translate into decent earnings, with fewer than 2% making the national minimum wage. 该条目发表于 Nature Food，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s43016-026-01342-z>

综合顶刊

40. Yellow glass shows insect wings at their best

(黄色玻璃让昆虫翅膀结构呈现最佳观测效果)

Nature | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Nature, Published online: 21 April 2026; doi:10.1038/d41586-026-00915-z An experiment to preserve the vivid colour of insect wings in displays and a disagreement about peer review in technology, in our weekly dip into Nature's archive.

中文摘要

研究围绕“黄色玻璃让昆虫翅膀结构呈现最佳观测效果”展开。英文摘要要点为：An experiment to preserve the vivid colour of insect wings in displays and a disagreement about peer review in technology, in our weekly dip into Nature's archive. 该条目发表于 Nature，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于综合顶刊方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/d41586-026-00915-z>

生态学核心

41. Fast range shifts

(物种分布范围的快速迁移)

Nature Ecology & Evolution | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Nature Ecology & Evolution, Published online: 21 April 2026; doi:10.1038/s41559-026-03072-8Fast range shifts

中文摘要

研究围绕“物种分布范围的快速迁移”展开。英文摘要要点为：Fast range shifts 该条目发表于 Nature Ecology & Evolution，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41559-026-03072-8>

环境与气候

42. [ASAP] Free Amino Acids in Marine Aerosols during Four Cruising Campaigns over the South China Sea: Importance of Biomass Burning and Atmospheric Oxidation

(南海航次海洋气溶胶游离氨基酸：生物质燃烧与大气氧化的重要作用)

Environmental Science & Technology | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c18403

中文摘要

研究围绕“南海航次海洋气溶胶游离氨基酸：生物质燃烧与大气氧化的重要作用”展开。英文摘要要点为：Environmental Science & TechnologyDOI: 10.1021/acs.est.5c18403 该条目发表于 Environmental Science & Technology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1021/acs.est.5c18403>

环境与气候

43. [ASAP] Fe(III) Oxide Reduction Bypassing Outer-Surface Cytochromes in a Marine Respiratory Anaerobe
(海洋呼吸厌氧菌绕过外表面细胞色素还原三价铁氧化物)

Environmental Science & Technology | 2026-04-21 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.6c04532

中文摘要

研究围绕“海洋呼吸厌氧菌绕过外表面细胞色素还原三价铁氧化物”展开。英文摘要要点为：Environmental Science & TechnologyDOI: 10.1021/acs.est.6c04532 该条目发表于 Environmental Science & Technology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

生态学核心

44. Estimating the Impacts of Future Extreme Heat on Dryland Threatened Mammals: An Australian Case Study
(估算未来极端高温对于干旱区受威胁哺乳动物的影响：澳大利亚案例)

Global Change Biology | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“估算未来极端高温对于干旱区受威胁哺乳动物的影响：澳大利亚案例”展开。英文摘要要点为：Estimating the Impacts of Future Extreme Heat on Dryland Threatened Mammals: An Australian Case Study 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70872>

植物与农业

45. Membrane remodelling mediates lipopeptide-induced immunity in Arabidopsis
(膜重塑介导拟南芥中脂肽诱导的免疫反应)

Nature Plants | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Nature Plants, Published online: 20 April 2026; doi:10.1038/s41477-026-02270-3 This work unveils a non-canonical lipid-driven mechanism for plant immune activation by a bacterial lipopeptide. Perception at the cell membrane leads to deformation, mechanosensing and early signalling and culminates in systemic resistance priming.

中文摘要

研究围绕“膜重塑介导拟南芥中脂肽诱导的免疫反应”展开。英文摘要要点为：This work unveils a non-canonical lipid-driven mechanism for plant immune activation by a bacterial lipopeptide. Perception at the cell membrane leads to deformation, mechanosensing and early signalling and culminates in systemic resistance priming. 该条目发表于 Nature Plants，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41477-026-02270-3>

土壤与生物地球化学

46. Phosphogypsum Doses and Subsoil Contamination: A Study on the Mobility of Rare Earth Elements, Uranium, and Thorium

(磷石膏施用量与底土污染：稀土元素、铀和钍迁移性研究)

European Journal of Soil Science | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“磷石膏施用量与底土污染：稀土元素、铀和钍迁移性研究”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science, 报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70325>

保护与生物多样性

47. Understanding drivers of coastal fishers' willingness to exit fishing in Zhejiang, China

(理解中国浙江沿海渔民退出捕捞意愿的驱动因素)

Conservation Biology | 2026-04-19 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“理解中国浙江沿海渔民退出捕捞意愿的驱动因素”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology, 报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

生态学核心

48. Grazing Stocking Rate and Rainfall Jointly Regulate Soil Methane Uptake in a Temperate Meadow Steppe, but Do Not Reverse Net Methane Emissions

(放牧载畜率与降雨共同调节温带草甸草原土壤甲烷吸收)

Global Change Biology | 2026-04-18 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“放牧载畜率与降雨共同调节温带草甸草原土壤甲烷吸收”展开。英文摘要要点为：Grazing Stocking Rate and Rainfall Jointly Regulate Soil Methane Uptake in a Temperate Meadow Steppe, but Do Not Reverse Net Methane Emissions 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70864>

生态学核心

49. Climate as the Main Driver of Large-Scale Genetic Patterns and Connectivity in the Expanding Golden Jackal

(气候是扩张中金豺大尺度遗传格局和连通性的主要驱动因子)

Global Change Biology | 2026-04-17 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“气候是扩张中金豺大尺度遗传格局和连通性的主要驱动因子”展开。英文摘要要点为：Climate as the Main Driver of Large-Scale Genetic Patterns and Connectivity in the Expanding Golden Jackal 该条目发表于 Global Change Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70879>

生态学核心

50. Decoupled Climatic Drivers of Tree and Ground-Layer Carbon Uptake in Mountain Ecosystems Around the World

(全球山地生态系统乔木层与地被层碳吸收的气候驱动相互脱耦)

Global Change Biology | 2026-04-17 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“全球山地生态系统乔木层与地被层碳吸收的气候驱动相互脱耦”展开。英文摘要要点为: Decoupled Climatic Drivers of Tree and Ground-Layer Carbon Uptake in Mountain Ecosystems Around the World 该条目发表于 Global Change Biology, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程, 提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70877>

生态学核心

51. Sexual Selection Associated With an Aggressive Male Phenotype Reduces Population Size and Hinders Population Recovery After Heat Stress

(攻击性雄性表型相关性选择降低种群规模并阻碍热胁迫后恢复)

Ecology Letters | 2026-04-17 | 【Latest】



图: Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“攻击性雄性表型相关性选择降低种群规模并阻碍热胁迫后恢复”展开。英文摘要要点为: Sexual Selection Associated With an Aggressive Male Phenotype Reduces Population Size and Hinders Population Recovery After Heat Stress 该条目发表于 Ecology Letters, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程, 提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70377>

海洋与水生

52. Urban water network upgrades improve quality and access to drinking water in the PAASIM matched cohort study in Beira, Mozambique

(莫桑比克贝拉 PAASIM 匹配队列显示城市供水网络升级改善饮水质量与可及性)

Nature Water | 2026-04-17 | 【Latest】



图： Graphical abstract / article figure source:

Abstract

Nature Water, Published online: 17 April 2026; doi:10.1038/s44221-026-00624-x Urban water systems in low-income cities often face contamination risks and unreliable supply, motivating evaluations of infrastructure upgrades. An assessment of the impact of water service line replacements on water quality and access in Beira, Mozambique, reveals reduced source water contamination and improved access.

中文摘要

研究围绕“莫桑比克贝拉 PAASIM 匹配队列显示城市供水网络升级改善饮水质量与可及性”展开。英文摘要要点为：Urban water systems in low-income cities often face contamination risks and unreliable supply, motivating evaluations of infrastructure upgrades. An assessment of the impact of water service line replacements on water quality and access in Beira, Mozambique, reveals reduced source water contamination and improved access. 该条目发表于 Nature Water，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s44221-026-00624-x>

遥感与模型

53. The Direct and Legacy Impacts of Droughts on Vegetation Growth in China

(干旱对中国植被生长的直接与滞后影响)

Journal of Geophysical Research - Biogeosciences | 2026-04-15 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“干旱对中国植被生长的直接与滞后影响”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于遥感与模型方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

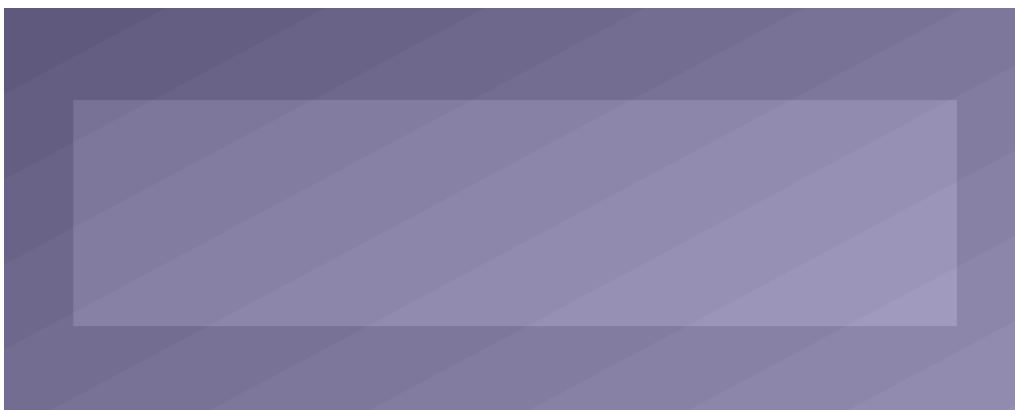
DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009199>

植物与农业

54. Early Postfire Regeneration Determines Long-Term Successional Trajectories in Temperate *Pinus densiflora* Forests of Korea

(火后早期更新决定韩国温带赤松林的长期演替轨迹)

Journal of Vegetation Science | 2026-04-13 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

中文摘要

研究围绕“火后早期更新决定韩国温带赤松林的长期演替轨迹”展开。英文摘要要点为：Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026. 该条目发表于 Journal of Vegetation Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70138>

生态学核心

55. Multiple Mechanisms Required to Predict Grass Community Composition

(预测草地群落组成需要多个机制共同支撑)

Ecology Letters | 2026-04-12 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“预测草地群落组成需要多个机制共同支撑”展开。英文摘要要点为：Multiple Mechanisms Required to Predict Grass Community Composition 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70358>

海洋与水生

56. A bioinspired hierarchical architecture for the high-yield recovery of industrial water vapour
(仿生层级结构实现工业水蒸气高产回收)

Nature Water | 2026-04-21 | 【Latest】



图： Graphical abstract / article figure source:

Abstract

Nature Water, Published online: 21 April 2026; doi:10.1038/s44221-026-00635-8 Industrial cooling towers release large quantities of water vapour that remains largely unutilized, despite its potential as a recoverable resource. This study demonstrates a passive, biomimetic multi-tier system to achieve high-rate water recovery with substantial large-scale savings potential.

中文摘要

研究围绕“仿生层级结构实现工业水蒸气高产回收”展开。英文摘要要点为：Industrial cooling towers release large quantities of water vapour that remains largely unutilized, despite its potential as a recoverable resource. This study demonstrates a passive, biomimetic multi-tier system to achieve high-rate water recovery with substantial large-scale savings potential. 该条目发表于 Nature Water，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s44221-026-00635-8>

环境与气候

57. [ASAP] Effects of Pesticide Mixtures and Environmental Factors on Benthic Diatom Communities: A Microcosm Approach

(农药混合物与环境因素对底栖硅藻群落的影响：微宇宙实验)

Environmental Science & Technology | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c14415

中文摘要

研究围绕“农药混合物与环境因素对底栖硅藻群落的影响：微宇宙实验”展开。英文摘要要点为：Environmental Science & TechnologyDOI: 10.1021/acs.est.5c14415 该条目发表于 Environmental Science & Technology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1021/acs.est.5c14415>

土壤与生物地球化学

58. Composite Soil Samples Reflect the Arithmetic Mean pH of the Individual Cores

(混合土样可反映单个土芯 pH 的算术平均值)

European Journal of Soil Science | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“混合土样可反映单个土芯 pH 的算术平均值”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究涉及碳、氮、水分或土壤生物过程，有助于改进生态系统碳汇和养分循环判断。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于土壤与生物地球化学方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70329>

遥感与模型

59. Spatiotemporal Modulation of Alkalinity and DIC Outwelling From Saltmarsh Porewater in New England's Largest Marsh Complex

(新英格兰最大盐沼复合体孔隙水碱度与溶解无机碳外输的时空调制)

Journal of Geophysical Research - Biogeosciences | 2026-04-20 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

中文摘要

研究围绕“新英格兰最大盐沼复合体孔隙水碱度与溶解无机碳外输的时空调制”展开。英文摘要要点为：Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026. 该条目发表于 Journal of Geophysical Research - Biogeosciences，报告中保留 DOI 与原文链接，便于后续核对全文细节。

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DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2026JG009711>

保护与生物多样性

60. Integrating traditional practices, livelihoods, and conservation with Indigenous-led furbearer camera trapping
(以原住民主导的毛皮动物红外相机监测整合传统实践、生计与保护)

Conservation Biology | 2026-04-19 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Conservation Biology, EarlyView.

中文摘要

研究围绕“以原住民主导的毛皮动物红外相机监测整合传统实践、生计与保护”展开。英文摘要要点为：Conservation Biology, EarlyView. 该条目发表于 Conservation Biology，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70290>

土壤与生物地球化学

61. Depth and Tillage Dependent Stoichiometry of C, N, S, and H in Ukrainian Mollisol
(乌克兰黑钙土碳氮硫氢化学计量随深度和耕作方式变化)

European Journal of Soil Science | 2026-04-19 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

中文摘要

研究围绕“乌克兰黑钙土碳氮硫氢化学计量随深度和耕作方式变化”展开。英文摘要要点为：European Journal of Soil Science, Volume 77, Issue 2, March–April 2026. 该条目发表于 European Journal of Soil Science, 报告中保留 DOI 与原文链接，便于后续核对全文细节。

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DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70335>

植物与农业

62. Conserved DNA architect couples chloroplast development to cell cycle in developing cotyledons

(保守 DNA 架构因子将子叶发育中的叶绿体发育与细胞周期耦合)

Nature Plants | 2026-04-17 | **【Latest】**



图：Graphical abstract / article figure source:

Abstract

Nature Plants, Published online: 17 April 2026; doi:10.1038/s41477-026-02280-1Light orchestrates cotyledon greening and growth, but the molecular link between chloroplast development and the cell cycle was unknown. Wang et al. identify RDE, a DNA-shaping protein that synchronizes these two processes in Arabidopsis.

中文摘要

研究围绕“保守 DNA 架构因子将子叶发育中的叶绿体发育与细胞周期耦合”展开。英文摘要要点为: Light orchestrates cotyledon greening and growth, but the molecular link between chloroplast development and the cell cycle was unknown. Wang et al. identify RDE, a DNA-shaping protein that synchronizes these two processes in Arabidopsis. 该条目发表于 Nature Plants, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫, 对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41477-026-02280-1>

植物与农业

63. Global food security rests on the Strait of Hormuz

(全球粮食安全依赖霍尔木兹海峡)

Nature Food | 2026-04-17 | **【Latest】**



图: Graphical abstract / article figure source:

Abstract

Nature Food, Published online: 17 April 2026; doi:10.1038/s43016-026-01340-1Global food security rests on the Strait of Hormuz

中文摘要

研究围绕“全球粮食安全依赖霍尔木兹海峡”展开。英文摘要要点为: Global food security rests on the Strait of Hormuz 该条目发表于 Nature Food, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫, 对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

植物与农业

64. Stress Dominance and Niche Conservatism in Epiphytic and Lithophytic Bromeliads Along a Coastal–Inland Gradient

(沿海到内陆梯度上附生和石生凤梨的胁迫主导与生态位保守性)

Journal of Vegetation Science | 2026-04-17 | **【Latest】**



图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

中文摘要

研究围绕“沿海到内陆梯度上附生和石生凤梨的胁迫主导与生态位保守性”展开。英文摘要要点为：Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026. 该条目发表于 Journal of Vegetation Science，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫，对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70137>

生态学核心

65. Group Size Dependent Selection for Cooperation Versus Freeloading in Collective Chemical Defence

(集体化学防御中合作与搭便车选择取决于群体大小)

Ecology Letters | 2026-04-16 | **【Latest】**



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究围绕“集体化学防御中合作与搭便车选择取决于群体大小”展开。英文摘要要点为：Group Size Dependent Selection for Cooperation Versus Freeloading in Collective Chemical Defence 该条目发表于 Ecology Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究围绕群落、种群、性状或生态系统过程，提供了可迁移的生态学机制认识。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于生态学核心方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70378>

植物与农业

66. Greening the scars

(让生态创伤重新变绿)

Nature Plants | 2026-04-15 | **【Latest】**



图：Graphical abstract / article figure source:

Abstract

Nature Plants, Published online: 15 April 2026; doi:10.1038/s41477-026-02287-8Greening the scars

中文摘要

研究围绕“让生态创伤重新变绿”展开。英文摘要要点为: Greening the scars 该条目发表于 Nature Plants, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫, 对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://doi.org/10.1038/s41477-026-02287-8>

植物与农业

67. Climatically Driven Stability in the Alpine/Moorland Boundary on Three Mountains in Southwest Lutruwita/Tasmania 1986–2020

(1986-2020 年塔斯马尼亚西南三座山地高山-沼原边界的气候驱动稳定性)

Journal of Vegetation Science | 2026-04-14 | **【Latest】**



图: Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

中文摘要

研究围绕“1986-2020 年塔斯马尼亚西南三座山地高山-沼原边界的气候驱动稳定性”展开。英文摘要要点为: Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026. 该条目发表于 Journal of Vegetation Science, 报告中保留 DOI 与原文链接, 便于后续核对全文细节。

深度解读

该研究连接植物功能、农业生产和环境胁迫, 对气候适应型农业与生态系统功能评估有启发。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言, 它补充了当周关于植物与农业方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性, 避免把单一区域或单一模型结论过度外推。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70136>

保护与生物多样性

68. Benthic diatoms and phytoplankton diversity in urban streams and ponds: charting a course for conservation (城市溪流与池塘中底栖硅藻和浮游植物多样性的保护启示)

Biodiversity and Conservation | 2026-04-17 | 【Latest】



图： Graphical abstract / article figure source:

Abstract

Benthic diatom and phytoplankton communities are key indicators of freshwater ecosystem health and change but remain understudied. A better understanding of their distribution and abundance drivers is crucial for effective biodiversity assessment and conservation. This study adopts a novel, multi-ecosystem approach to examine the seasonal (2019–2020) diversity of benthic diatom assemblages in urban streams and phytoplankton assemblages in ponds across Bristol (UK) and the local environmental factors shaping these communities. Among ponds, winter recorded significantly fewer phytoplankton taxa than summer and autumn. Among streams, patterns of benthic diatom beta diversity indicated higher turnover in autumn than in winter, and among pond habitats, turnover dominated in the summer and autumn.

中文摘要

研究围绕“城市溪流与池塘中底栖硅藻和浮游植物多样性的保护启示”展开。英文摘要要点为：Benthic diatom and phytoplankton communities are key indicators of freshwater ecosystem health and change but remain understudied. A better understanding of their distribution and abundance drivers is crucial for effective biodiversity assessment and conservation. This study adopts a novel, multi-ecosystem approach to examine the seasonal (2019–2020) diversity of benthic diatom assemblages in urban streams and phytoplankton assemblages in ponds across Bristol (UK) and the local environmental factors shaping these communities. Among ponds, winter recorded significantly fewer phytoplankton taxa than summer and autumn. Among streams, patterns of benthic diatom beta diversity indicated higher turnover in autumn than in winter, and among pond habitats, turnover dominated in the summer and autumn. 该条目发表于 Biodiversity and Conservation，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://link.springer.com/article/10.1007/s10531-026-03305-z>

环境与气候

69. Increasing extremity and accelerating expansion of heatwaves across Eurasia (欧亚大陆热浪极端性增强且扩张加速)

Environmental Research Letters | 2026-04-16 | 【Latest】



图： Graphical abstract / article figure source:

Abstract

Heatwaves have increased globally under climate warming. However, quantitative assessments regarding their spatial expansion and extremity evolution at the continental scale remain limited. In this study, we used multiple datasets and a unified framework to assess long-term changes in heatwave frequency, intensity and spatial clustering across 15 reference regions of the Intergovernmental Panel on Climate Change in Eurasia from 1950 to 2023. In Eurasia, heatwaves have shifted toward more extreme conditions since the 1990s. The spatial extent of rare events, which represent the most extreme conditions exceeding the 90th percentile of the 1991–2020 baseline period, expanded most rapidly. The annual mean area affected by rare heatwaves during 2014–2023 has tripled relative to the climatologic

中文摘要

研究围绕“欧亚大陆热浪极端性增强且扩张加速”展开。英文摘要要点为：Heatwaves have increased globally under climate warming. However, quantitative assessments regarding their spatial expansion and extremity evolution at the continental scale remain limited. In this study, we used multiple datasets and a unified framework to assess long-term changes in heatwave frequency, intensity and spatial clustering across 15 reference regions of the Intergovernmental Panel on Climate Change in Eurasia from 1950 to 2023. In Eurasia, heatwaves have shifted toward more extreme conditions since the 1990s. The spatial extent of rare events, which represent the most extreme conditions exceeding the 90th percentile of the 1991–2020 baseline period, expanded most rapidly. The annual mean area affected by rare heatwaves during 2014–2023 has tripled relative to the climatologic 该条目发表于 Environmental Research Letters，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究与气候变化、极端事件或环境治理直接相关，能帮助判断生态系统和人类社会面对的复合风险。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于环境与气候方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae5990>

保护与生物多样性

70. Unveiling shifting frontiers of tropical dry forests depends on consistently accurate delineation of their distribution: remarks about Rivas et al. (2025) and others

(揭示热带干旱森林变动边界依赖持续准确的分布划定)

Biodiversity and Conservation | 2026-04-17 | 【Latest】



图： Graphical abstract / article figure source:

Abstract

Recently in Biodiversity and Conservation, it was asserted that 9.64% and 5.96% of tropical dry forests (TDFs) in South America and Africa were lost between 2000 and 2020. Arriving at such results depends on how TDFs are defined and how their global distribution is mapped. A striking feature of recent maps of TDFs is the substantial overlap between them and maps of tropical grassy ecosystems (TGEs). Alarming, both types of maps are generally based on just two earlier maps of global terrestrial ecosystems. If extensive areas of global TGEs are being misclassified as TDFs, then there are problems because policies relevant to forests, e.g., fire suppression and ‘reforestation’ are not applicable to TGEs. Mapping of TGEs typically relies on ecologically functional criteria such as the distri

中文摘要

研究围绕“揭示热带干旱森林变动边界依赖持续准确的分布划定”展开。英文摘要要点为：Recently in Biodiversity and Conservation, it was asserted that 9.64% and 5.96% of tropical dry forests (TDFs) in South America and Africa were lost between 2000 and 2020. Arriving at such results depends on how TDFs are defined and how their global distribution is mapped. A striking feature of recent maps of TDFs is the substantial overlap between them and maps of tropical grassy ecosystems (TGEs). Alarming, both types of maps are generally based on just two earlier maps of global terrestrial ecosystems. If extensive areas of global TGEs are being misclassified as TDFs, then there are problems because policies relevant to forests, e.g., fire suppression and ‘reforestation’ are not applicable to TGEs. Mapping of TGEs typically relies on ecologically functional criteria such as the distri 该条目发表于 Biodiversity and Conservation，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://link.springer.com/article/10.1007/s10531-026-03343-7>

保护与生物多样性

71. Robots for ecological monitoring: a review

(生态监测机器人研究综述)

Biodiversity and Conservation | 2026-04-16 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Technologies for biodiversity assessment are crucial for addressing the global biodiversity crisis, and ecological monitoring is seeing increased use of robotics. Interdisciplinary working will be critical in finding the best robotic solutions for monitoring. As a team of ecologists and robotics scientists, we conducted a systematic literature search for robots in ecological monitoring and reviewed their ecological monitoring applications and technological capabilities, such as level of autonomy, operation time and adaptability. We identified 67 relevant papers reporting individual robots or robot teams. The number of robots developed for ecological monitoring has grown rapidly in recent years, targeting a wide range of taxa or communities. Image or video capture were the most common types

中文摘要

研究围绕“生态监测机器人研究综述”展开。英文摘要要点为：Technologies for biodiversity assessment are crucial for addressing the global biodiversity crisis, and ecological monitoring is seeing increased use of robotics. Interdisciplinary working will be critical in finding the best robotic solutions for monitoring. As a team of ecologists and robotics scientists, we conducted a systematic literature search for robots in ecological monitoring and reviewed their ecological monitoring applications and technological capabilities, such as level of autonomy, operation time and adaptability. We identified 67 relevant papers reporting individual robots or robot teams. The number of robots developed for ecological monitoring has grown rapidly in recent years, targeting a wide range of taxa or communities. Image or video capture were the most common types 该条目发表于 Biodiversity and Conservation, 报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究关注物种、栖息地或保护成效，强调生物多样性风险需要与土地利用和气候变化共同评估。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于保护与生物多样性方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://link.springer.com/article/10.1007/s10531-026-03339-3>

海洋与水生

72. Fine-scale aquatic eDNA sampling reveals significant within- and across-site variation during the expected breeding season of *Cryptobranchus alleganiensis alleganiensis*



图：Graphical abstract / article figure source:

Abstract

Environmental DNA (eDNA) analysis is a powerful tool for monitoring wildlife populations. While seasonal peaks in eDNA concentration during breeding seasons of target aquatic species are well documented, fine-scale temporal changes in eDNA are yet to be extensively examined. We used *C. a. alleganiensis*, an aquatic salamander whose breeding season is characterized by high eDNA concentrations, as a model species to test three hypotheses: eDNA concentration would remain high throughout the expected breeding season, eDNA signatures would reflect synchronous breeding among local populations, and eDNA concentration changes would correspond to seasonal stream-temperature changes. We sampled five tributaries of the Susquehanna River in Pennsylvania, USA, daily from August to October to examine how

中文摘要

研究围绕“细尺度水生 eDNA 采样揭示大鲵预期繁殖季站点内外显著变异”展开。英文摘要要点为：Environmental DNA (eDNA) analysis is a powerful tool for monitoring wildlife populations. While seasonal peaks in eDNA concentration during breeding seasons of target aquatic species are well documented, fine-scale temporal changes in eDNA are yet to be extensively examined. We used *C. a. alleganiensis*, an aquatic salamander whose breeding season is characterized by high eDNA concentrations, as a model species to test three hypotheses: eDNA concentration would remain high throughout the expected breeding season, eDNA signatures would reflect synchronous breeding among local populations, and eDNA concentration changes would correspond to seasonal stream-temperature changes. We sampled five tributaries of the Susquehanna River in Pennsylvania, USA, daily from August to October to examine how 该条目发表于 Hydrobiologia，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://link.springer.com/article/10.1007/s10750-026-06186-z>

海洋与水生

73. Sixty years of artificial lake heating impact *Daphnia* fitness but not parasite resistance

(六十年人工湖泊增温影响水蚤适合度但不影响抗寄生能力)



图： Graphical abstract / article figure source:

Abstract

The impacts of climate change are expected to be complex and severe, with rising temperatures potentially exacerbating the prevalence of infectious diseases. Aquatic ectotherms, such as *Daphnia*, are particularly vulnerable to temperature fluctuations, which can significantly impact their fitness and disease resistance. Using a unique system of lakes, some of which have been artificially heated for over 60 years, we investigated how elevated temperatures have affected fitness and parasite resistance in *Daphnia longispina*. Individuals from heated lakes as well as from nearby control lakes were experimentally exposed to the allopatric fungal parasite *Metschnikowia bicuspidata* at two temperatures (18 °C and 22 °C). We found that *Daphnia* exhibited high resistance to *Metschnikowia*, regardless of

中文摘要

研究围绕“六十年人工湖泊增温影响水蚤适合度但不影响抗寄生能力”展开。英文摘要要点为：The impacts of climate change are expected to be complex and severe, with rising temperatures potentially exacerbating the prevalence of infectious diseases. Aquatic ectotherms, such as *Daphnia*, are particularly vulnerable to temperature fluctuations, which can significantly impact their fitness and disease resistance. Using a unique system of lakes, some of which have been artificially heated for over 60 years, we investigated how elevated temperatures have affected fitness and parasite resistance in *Daphnia longispina*. Individuals from heated lakes as well as from nearby control lakes were experimentally exposed to the allopatric fungal parasite *Metschnikowia bicuspidata* at two temperatures (18 °C and 22 °C). We found that *Daphnia* exhibited high resistance to *Metschnikowia*, regardless of 该条目发表于 *Hydrobiologia*，报告中保留 DOI 与原文链接，便于后续核对全文细节。

深度解读

该研究把水文过程、海洋生态或淡水生态响应放在核心位置，对流域和近岸管理具有参考价值。其重要性在于把具体观测或模型结果转化为可检验的机制判断。对 2026-04-25 这期速览而言，它补充了当周关于海洋与水生方向的前沿证据。后续使用时应结合全文方法、空间尺度和不确定性，避免把单一区域或单一模型结论过度外推。

DOI: <https://link.springer.com/article/10.1007/s10750-025-06106-7>